

ister the native host manifest with Chrome. In return, the assistant gains stable, full-duplex communication without exposing a network listener on the public internet; the host binds an optional local HTTP endpoint (`127.0.0.1:8080`) for development dashboards only.

3.5 Python Native Host

The host entry point is `testing/host/host.py`. On startup it loads configuration from `config.py` (environment variables and `.env`), initialises data directories under `testing/host/data/`, and starts the native messaging I/O loop via the dispatcher module. Incoming messages are classified by type and routed to specialised modules: notebook ingestion (`notebook_data_handler.py`), streaming chat (`streaming.py`), and tool execution (`tool_registry.py` and `agentic_batch_executor.py`).

3.5.1 Prompt assembly and chat modes

Prompt templates reside under `testing/host/prompts/`, with mode-specific files for Ask, Code, and Agentic behaviour. The module `prompt_engineering.py` composes the message list sent to the [API](#): a base notebook-assistant system prompt, Jupyter structure guidance, mode instructions, a packed notebook context slice, and recent chat history subject to token budgets (`MAX_INPUT_TOKENS` and `MAX_NOTEBOOK_CONTEXT_CHARS`). Context packing can operate in `full` or `intent` mode; the default favours complete cell lists when token limits allow, supporting faithful positional references in tool arguments. Figure 3.5 depicts the context-packing and notebook-intelligence pipeline.

Chat history is keyed by a stable notebook identity resolved from the Kaggle kernel id when available (`kaggle:kernel:<id>`) or from a normalised URL fallback (`notebook_identity.py`). This key aligns snapshot filenames, SQLite session storage, and tool `url` arguments.

3.5.2 Tool registry and agentic execution contract

Tools are registered in `tool_registry.py` with JSON-schema argument validation. They fall into two classes:

- **Local read tools** execute entirely on the host against persisted JSON snapshots. Examples include `notebook_list_cells`, `notebook_get_cell`, and `notebook_snapshot_status`.